Computer Architecture – CO1 Assessment Tool 1

Question 1

1. A programmer wants to understand how data is transferred between registers during program execution in an 8085 microprocessor. To observe register operations during program execution, write and execute an 8085 program that performs data transfer and arithmetic operations using multiple registers. Observe the contents of the Accumulator and the general-purpose registers (B, C, D, E, H, and L) after each instruction. Analyse how register values change during execution and determine the role of each register in completing the program.

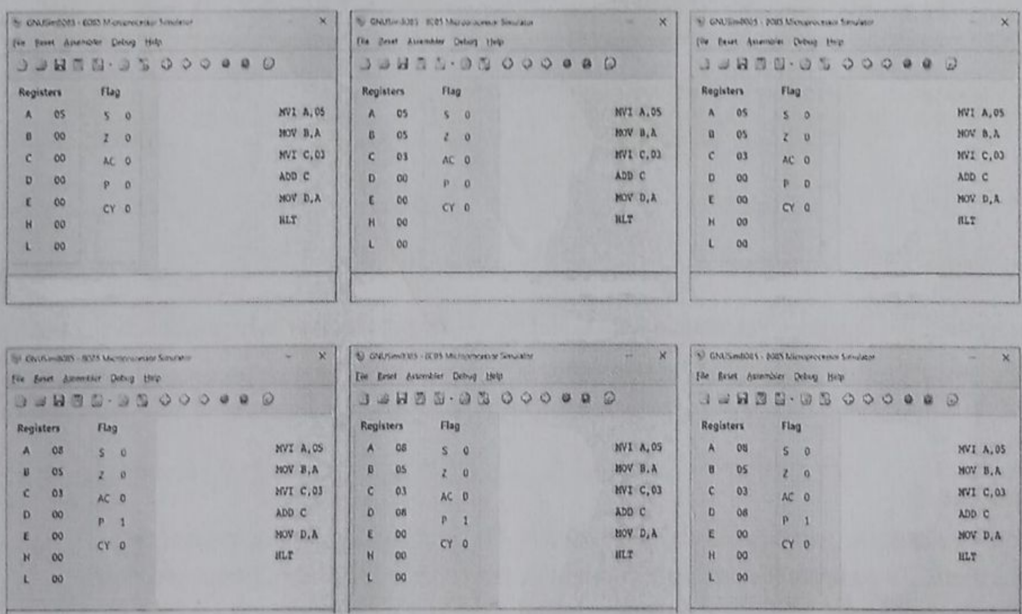
8085 Register Observation

Program used:

```
MVI A,05
MOV B,A
MVI C,03
ADD C
MOV D,A
HLT
```

Observation

- A loaded with 05.
- B receives value from A.
- C loaded with 03.
- ADD C -> A becomes 08.
- D stores the result 08.

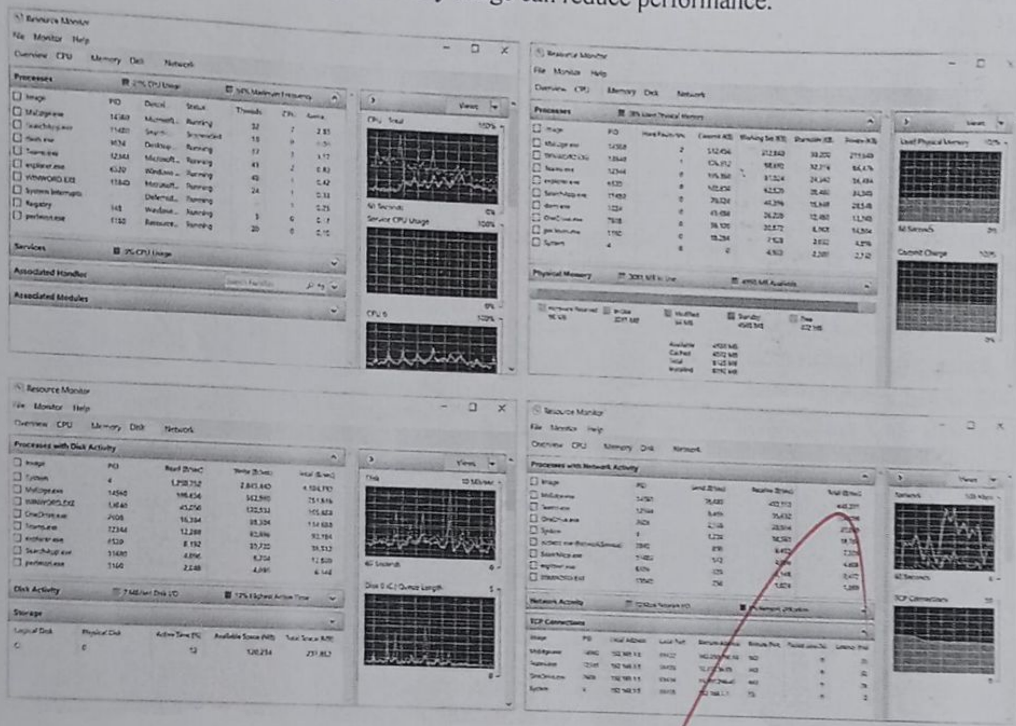


Question 2

A user notices that a computer responds slowly while attending an online discussion, editing a document, and downloading files simultaneously. To analyse the system behaviour under multitasking, observe the Memory usage, Disk activity and Network activity while performing these tasks. Analyse how the CPU, memory, and I/O devices cooperate during multitasking, and determine which hardware resource is most likely to become the performance bottleneck.

Short analysis:

- CPU executes multiple tasks.
- Memory stores active programs.
- Disk reads/writes files.
- Network transfers online meeting and downloads.
- Bottleneck: Slow HDD or high memory usage can reduce performance.



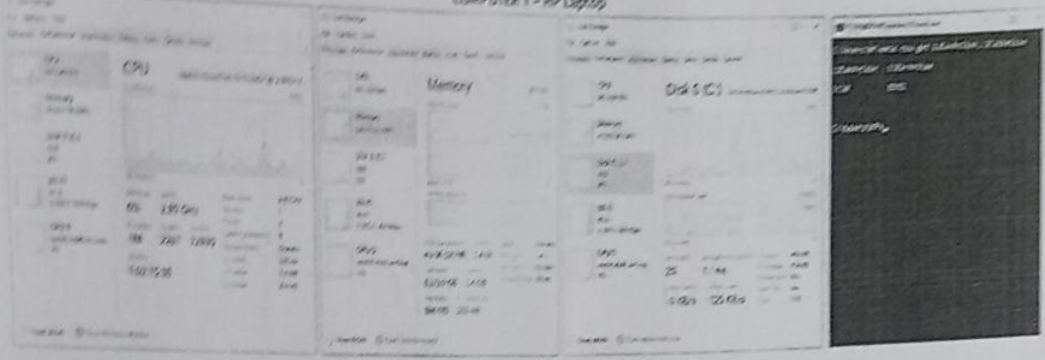
Conclusion:

During multitasking, the CPU, memory, disk, and network work together, with the disk being the most likely performance bottleneck due to heavy read/write operations.

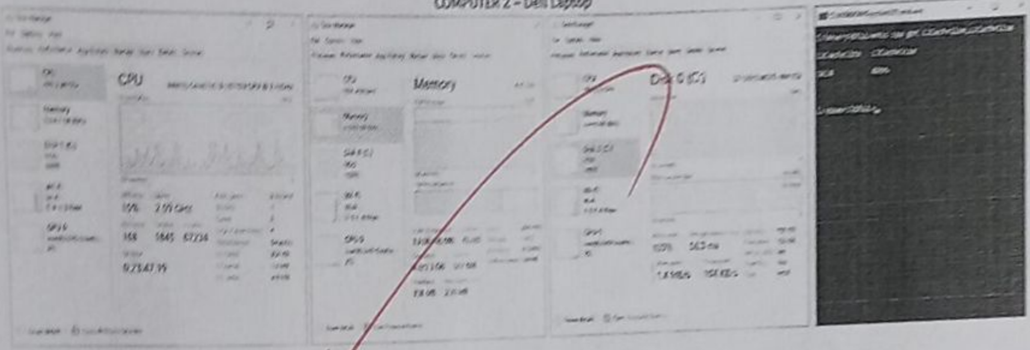
Question 3

A software engineer wants to identify the most suitable computer for running engineering applications. To compare the hardware capabilities, observe the Processor Generation, Base Speed, Current Speed, Virtualization Status, Installed RAM, Cache Memory (L1/L2/L3 if available), and Storage Type (SSD/HDD) for two computers. Analyse the observed specifications to determine which system is expected to execute programs more efficiently, and justify your conclusion by relating these hardware characteristics to processor performance.

COMPUTER 1 – HP Laptop



COMPUTER 2 – Dell Laptop



Specification	Computer 1 (HP Laptop)	Computer 2 (Dell Laptop)
Processor Generation	11th Gen Intel Core i5	10th Gen Intel Core i3
Base Speed	2.40 GHz	2.10 GHz
Current Speed	2.89 GHz	2.59 GHz
Virtualization	Enabled	Disabled
Installed RAM	8 GB	4 GB
L1/L2/L3 Cache	L1: 320 KB L2: 5.0 MB L3: 8.0 MB	L1: 256 KB L2: 1.0 MB L3: 4.0 MB
Storage (SSD/HDD)	SSD (NVMe 512 GB)	HDD (1 TB)

Short Analysis

- Processor: Computer 1 has a newer 11th Gen Intel Core i5, while Computer 2 has a 10th Gen Intel Core i3.
- Speed: Computer 1 has a higher base and current clock speed.
- Virtualization: Enabled on Computer 1, making it suitable for virtual machines and engineering software.
- RAM: Computer 1 has 8 GB, which is better for multitasking than the 4 GB in Computer 2.
- Cache: Computer 1 has larger L1, L2, and L3 caches, improving CPU performance.
- Storage: Computer 1 uses an SSD, which is much faster than the HDD in Computer 2.

Conclusion

Computer 1 (HP Laptop) is expected to execute engineering applications more efficiently because it has:

- Newer processor generation
- Higher clock speed
- Larger cache memory
- More RAM
- Virtualization enabled
- SSD storage

These features improve processing speed, multitasking, and overall system performance.